FIRST MOMENT OF CENTRAL L-DERIVATIVES FOR QUADRATIC TWISTS REPRESENTED AS SUMS OF TWO SQUARES

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Allowed standard prerequisites (editorial).

- Modular L-functions, approximate functional equations, Voronoi summation and quadratic-twist moment estimates. Complete original bibliography and author-local derivations retained.

1. Introduction and Main Results

Let E be an elliptic curve over $\mathbb{Q}$. The Birch and Swinnerton-Dyer (BSD) conjecture dictates that the analytic and algebraic ranks of E are actually the same invariant. Recent years have seen substantial progress towards BSD, and in particular for small ranks. The first such breakthrough was due to Gross–Zagier [5] and Kolyvagin [12] which together establish that if E has analytic rank 0 (respectively 1), then E has algebraic rank 0 (respectively 1) provided we know something about the twist of E by a quadratic character χ_d, which we denote by E^d. In particular, what we want to know is that if there exists some d such that E^d has analytic rank 0 or 1, i.e. if the normalized L-function $L(s, E\otimes\chi_d)$ doesn't vanish or has a simple pole at $s = 1/2$. Such statements were first proven by [24, 17, 2]. The study of twists of E has thus become one of the central areas in number theory since then, and other reasons include their relations to Fourier coefficients of automorphic forms, see for instance [23, 11].

Studying E^d when d runs over twists in thinner families is a harder question. We mention for instance the result of Coates–Li–Tian–Zhai in [3], which proves the existence of infinitely many twists d which are products of exactly k prime factors such that E^d has rank 0 (or 1), under certain conditions on E.

In this paper, we consider twists which are sums of two squares. Such a family of twists is non-linear and information on ranks of E^d for these d's would give us information about the elliptic surface $(1 + t^2)y^2 = f(x)$, where $f(x)$ is a cubic polynomial, via the Shioda–Tate formula for instance (cf. [20]); in particular information on its Picard number and generic rank. This family was also considered by Munshi in [15], but results on ranks of E^d don't follow from his methods. In the case E has Complex Multiplication (CM) however, Munshi [16] was able to get results about ranks. We will show such results in the general non-CM case, and thus extend the work of Munshi in [15]. We first start by introducing some definitions and notations.

Let $f \in S_\kappa(N)$ be a normalized Hecke eigenform of even weight κ and level N. Denote by $\lambda_f(n)$ the nth normalized Fourier coefficient of f, i.e.

$$f(z) = e(z) + \sum_{n\geq 2} \lambda_f(n) n^{\frac{\kappa-1}{2}} e(nz).$$

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Its L-function is defined by

$$L(s,f)=\sum_{n=1}^{\infty}\frac{\lambda_f(n)}{n^s}=\prod_{p\nmid N}\frac{1}{1-\lambda_f(p)p^{-s}+p^{-2s}}\prod_{p|N}\frac{1}{1-\lambda_f(p)p^{-s}}$$

for $\mathrm{Re}(s)>1$. Define the completed L-function as

$$\Lambda(f,s)=N^{s/2}(2\pi)^{-s}\Gamma\left(s+\frac{\kappa-1}{2}\right)L(s,f).$$

This satisfies the functional equation

$$\Lambda(s,f)=i^{\kappa}\eta\,\Lambda(1-s,f),$$

where η is the eigenvalue of the Fricke involution W_N acting on f. We denote by $w(f)=i^{\kappa}\eta$ the root number of f. The twisted L-function of f by a fundamental discriminant d with $(d,N)=1$ will satisfy

$$L(s,f\otimes\chi_d)=\sum_{n=1}^{\infty}\frac{\lambda_f(n)\chi_d(n)}{n^s}=\prod_{p\nmid dN}\frac{1}{1-\lambda_f(p)\chi_d(p)p^{-s}+p^{-2s}}\prod_{p|N}\frac{1}{1-\lambda_f(p)\chi_d(p)p^{-s}},$$

and if we set

$$\Lambda(s,f\otimes\chi_d)=(|d|N)^{s/2}(2\pi)^{-s}\Gamma\left(s+\frac{\kappa-1}{2}\right)L(s,f\otimes\chi_d),$$

then we have the functional equation

$$\Lambda(s,f\otimes\chi_d)=i^{\kappa}\eta\chi_d(-N)\Lambda(1-s,f\otimes\chi_d).$$

Note that the level of the twisted modular form $f\otimes\chi_d$ becomes larger and in this case equals Nd^2.

Let d be an odd, positive, squarefree (SF) integer. Then $\chi_{8d}(n)$ is a real primitive character with conductor $8d$ and satisfies $\chi_{8d}(-1)=1$. Let

$$r(d)=\#\{(a,b)\in\mathbb{Z}^2\setminus(0,0):d=a^2+b^2\}$$

be the representation function for sums of two squares. We prove the following result.

Theorem 1. *Let $f\in S_{\kappa}(\Gamma_0(N))$ be a normalized eigenform, and ψ be a smooth function with compact support in $(0,1]$. We have*

$$\sum_{\substack{d\ SF\\(d,2N)=1\\w(f\otimes\chi_{8d})=-1}}r(d)L'(1/2,f\otimes\chi_{8d})\psi\left(\frac{8d}{X}\right)=\mathcal{C}X\log X+O(X(\log X)^{1/2}(\log\log X)^3),$$

with

$$\mathcal{C}=\frac{1}{2^7}\hat{\psi}(0)\prod_{p>2}\left(1-\frac{\rho(p^2)}{p^4}\right)L(1,sym^2f)\Big[A(0,N,1)-w(f)A(0,N,N)\Big],$$

where $\hat{\psi}$ is the Fourier transform of ψ, the function ρ is defined in (3), $L(1,sym^2f)$ is defined in (6), and $A(0,N,N')$ is defined in (5) for $N'\in\{1,N\}$. Note that $\mathcal{C}=0$ if and only if $w(f)=1$ and N is a perfect square.

Such asymptotics for higher derivatives of the completed L-function $\Lambda^{(\ell)}(1/2,f\otimes\chi_{8d})$ were obtained in [15], but Theorem 1 is one of the first results that deals with lower derivatives in such a family, and with the L-function itself rather than with the completed one.

A direct consequence of Theorem 1 is the following.

Corollary 2. *Let E be an elliptic curve over $\mathbb{Q}$ of odd conductor N. If $w(E) = 1$, then assume also that N is not a perfect square. There are infinitely many fundamental discriminants $d = a^2 + b^2$ such that E^d has algebraic rank* 1.

Remark 3. *Theorem 1 can be extended to any primitive positive-definite binary quadratic form $Q(x, y) = ax^2 + bxy + cy^2$, i.e. to study*[1]

$$\sum_{\substack{d\equiv 1 \bmod 4\\ d\ SF\\ (d,N)=1\\ w(f\otimes\chi_d)=-1}} r_Q(d)L'(1/2, f\otimes\chi_d)\psi\left(\frac{d}{X}\right),$$

and

$$\sum_{\substack{d\ SF\\ (d,2N)=1\\ w(f\otimes\chi_{8d})=-1}} r_Q(d)L'(1/2, f\otimes\chi_{8d})\psi\left(\frac{8d}{X}\right),$$

where

$$r_Q(d) := \{(a, b) \in \mathbb{Z}^2 \setminus (0, 0) : d = Q(a, b)\}.$$

Specifically, for Proposition 8, one has a general Voronoi summation formula for $r_Q(d)$ (see Remark 9). The additional modifications will be in the other results of Section 3, where one has to be careful incorporating $Q(x, y)$ (and, for the former sum, the case of even entries coming from the L-function[2]*) into the study of the general two-dimensional Gauss sums defined in Remark 9.*

The first moment of L' but over the full family (i.e. without the $r(d)$ weight) was obtained in many papers, cf. [17, 9]. The first moment of L over the full family instead of L' was established in [9, 19, 21]. Asymptotics for second moments over the full family of twists can be found under GRH in [22, 18], and unconditionally in the work of Li [14]. Following Li's ideas, unconditional mixed second moments were further done in [13, 25, 6, 10]. For analogous moment results over function fields, see [1].

1.1 Strategy of the Proof

The proof follows the general strategy as that of Soundararajan and Young [22]. We begin with the approximate functional equation (Lemma 5),

$$L'(1/2, f\otimes\chi_d) \approx \sum_{n\geq 1} \frac{\lambda_f(n)\chi_d(n)}{n^{1/2}} W\left(\frac{n}{d}\right).$$

We then split the sum over n into a short range and a long range. For the short range, we introduce the weight $W(n/U)$, where

$$U = \frac{X}{(\log X)^A},$$

with $A > 0$ sufficiently large. The long range is then captured by the complementary weight

$$W\left(\frac{n}{d}\right) - W\left(\frac{n}{U}\right).$$

[1] In the former sum, the main term vanishes if the binary quadratic form $Q(u, v)$ is congruent to 1 and 5 $\pmod 8$ equally often as (u, v) ranges over residue classes modulo 8.

[2] Writing $n = 2^\alpha n'$ with n' odd, isolates the powers of 2, which imposes additional restrictions on the Gauss sums. The resulting sum over α will converge, and the argument proceeds similarly as in the odd case treated in the paper.

For the short range, after temporarily ignoring the squarefree and coprimality conditions, we are led to study sums of the form

$$\sum_{\substack{d\geq 1\\ w(f\otimes\chi_{8d})=-1}} r(d)\sum_{n\geq 1}\frac{\lambda_f(n)\chi_{8d}(n)}{n^{1/2}}h(d,n),$$

where $h(x,y)$ is a smooth function, compactly supported in the x-variable. After interchanging the order of summation, we apply the Voronoi summation formula of $r(d)$ (Proposition 8). Since we employ Voronoi summation rather than Poisson summation, the resulting dual sum has expected length squared; consequently, the analysis resembles that of a second moment rather than a first moment.

This reduces the problem to estimating sums of the form

$$\sum_{(k_1,k_2)\in\mathbb{Z}^2}(-1)^{k_1+k_2}G_{(k_1,k_2)}(n)\check{h}\left(\frac{(k_1^2+k_2^2)X}{2n^2},n\right),$$

where

$$G_{(k_1,k_2)}(n)=\sum_{b_1,b_2 \bmod n}\left(\frac{b_1^2+b_2^2}{n}\right)e\left(\frac{k_1b_1}{n}+\frac{k_2b_2}{n}\right)$$

is a two-dimensional Gauss sum, and $\check{h}(\xi,y)$ denotes the Bessel transform of $h(x,y)$ with respect to the x-variable (see (1)).

The zero frequency $(k_1,k_2)=(0,0)$ produces the main term, following an argument analogous to that of Soundararajan and Young. For the nonzero frequencies, we invoke Mellin inversion to $\check{h}$, expressing it as a double contour integral. After a suitable factorization of k_1 and k_2, the resulting expression can be recast in terms of sums of central twists $L(1/2+it,f\otimes\chi_d)$. We then employ Li's estimate [14] to obtain the required bounds.

Finally, the contribution from the long range is controlled using the work of Zhou [25, Proposition 4.3] via an application of Cauchy–Schwarz.

Remark 4. *Concurrently with the completion of this paper, a paper by Huang and Wei* [7] *just appeared where they establish a similar result for the convolutions* $1*\chi_D$*, which in particular recovers an analogue of Theorem 1. Their work does not consider the family subject to the root number condition on the twist; however, with their restriction on the discriminants, they can get an analogue of Corollary 2 subject to a global root number condition on the elliptic curve. Our approach was developed independently and differs substantially in its treatment of the frequency side, which constitutes the principal point of departure; the treatment of the error terms in the remainder of the argument is similar in spirit. More precisely, Huang and Wei employ Mellin inversion to study averages of Dirichlet* L*-functions twisted by the Fourier coefficients* $\lambda_f(n)$*. By contrast, our argument follows the classical Poisson summation philosophy, specifically through a Voronoi summation formula, which leads to the analysis of certain two-dimensional Gauss sums. The Voronoi summation formula established in Proposition 8 may be of independent interest, together with the understanding of the properties of these relevant Gauss sums which is the content of Section 3.*

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1.2 NOTATION

We use the usual Vinogradov asymptotic notation. Throughout the paper, $r(d)$ is the representation function for the sum of two squares, and $J_0(x)$ is the usual Bessel function of the first kind. We use $\left(\frac{\cdot}{\cdot}\right)$ for the Kronecker symbol, and $e(x) = e^{2i\pi x}$. We use $\tilde{h}$ for the Mellin transform, $\hat{f}$ for the Fourier transform, and $\check{g}$ for the Bessel transform as defined in (1). Further, throughout the paper, we use $\tau(n)$ for the divisor function. We will usually write $\tau(n)^{O(1)}$ to denote an unspecified fixed power of the divisor function. In certain parts of the paper, we use τ_p to denote the usual Gauss sum $\bmod p$, namely

$$\tau_p = \sum_{x \bmod p} \left(\frac{x}{p}\right) e\left(\frac{x}{p}\right).$$

2. PRELIMINARIES

In this section, we will collect certain Lemmas which we will find useful throughout.

Lemma 5. *Let $f \in S_\kappa(\Gamma_0(N))$ be a normalized eigenform with root number $w(f) = i^\kappa \eta$. We have that*

$$L'(1/2, f \otimes \chi_d) = \sum_{n\geq 1} \frac{\lambda_f(n)\chi_d(n)}{n^{1/2}} W\left(\frac{n}{|d|}\right) + i^\kappa \eta \chi_d(-N) \sum_{n\geq 1} \frac{\lambda_f(n)\chi_d(n)}{n^{1/2}} W\left(\frac{n}{|d|}\right),$$

where $W(y)$ is a smooth function defined by

$$W(y) = \frac{1}{2\pi i} \int_{(3)} \frac{\Gamma(u+\kappa/2)}{\Gamma(\kappa/2)} \left(\frac{2\pi y}{\sqrt{N}}\right)^{-u} \frac{du}{u^2}.$$

Proof. This follows from the classical proof for the L-function $L(s, f)$ in [8, Theorem 5.3]. We refer the reader to [18, Lemma 1] for a complete proof for the derivative $L'(s, f)$. □

The function W and its derivatives $W^{(j)}$ satisfy the following bounds.

Lemma 6. *For any $y \in \mathbb{R}$ and $j \in \mathbb{Z}_{\geq 1}$, we have that*

$$W^{(j)}(y) \ll_{j,B} \frac{1}{y^j} \left(\log \frac{\kappa N}{y}\right) \left(1 + \frac{y}{\kappa\sqrt{N}}\right)^{-B}$$

for any $B > 0$.

Proof. The bounds follow in the same way as in the proof of [8, Proposition 5.4], taking into consideration the extra y^{-1} factor in the defining integral. □

One of the important consequences of Li's work [14] is the following unconditional bound for the second moment of the twisted L-function.

Lemma 7. *Let $f \in S_\kappa(\Gamma_0(N))$. Then we have the bound*

$$\sum_{0<d\leq X} \left| L\left(\frac{1}{2} + it, f \otimes \chi_d\right)\right|^2 \ll_f (1+|t|)^2 X \log X.$$

Proof. This follows from [14, Proof of Theorem 1.1]. □

3. Voronoi Summation for Sums of Two Squares and Two-Dimensional Gauss Sums

We first start by proving a Voronoi summation formula that incorporates the representation function $r(d)$.

Proposition 8 (Voronoi Summation Formula)**.** *Let F be a smooth function with compact support on the positive real numbers, and n an odd integer. Let also M be a positive integer coprime to $2n$. Then*

$$\sum_{\substack{(d,2)=1\\ d\equiv 0 \bmod M}} r(d)\left(\frac{d}{n}\right) F\left(\frac{d}{Z}\right) = \frac{Z}{4M^2n^2}\left(\frac{2}{n}\right)\sum_{(k_1,k_2)\in\mathbb{Z}^2}(-1)^{k_1+k_2}G_{(k_1,k_2)}(M,n)\check{F}\left(\frac{(k_1^2+k_2^2)Z}{2M^2n^2}\right)$$

where

$$G_{(k_1,k_2)}(M,n) := \sum_{\substack{b_1,b_2 \bmod Mn\\ b_1^2+b_2^2\equiv 0 \bmod M}}\left(\frac{b_1^2+b_2^2}{n}\right) e\left(\frac{k_1b_1}{Mn}+\frac{k_2b_2}{Mn}\right),$$

and

$$\check{F}(u) = 2\pi\int_0^\infty F(t)J_0(2\pi\sqrt{ut})dt, \tag{1}$$

is the Bessel transform of F.

Proof. We start by getting rid of the $(d,2)=1$ condition;

$$\sum_{\substack{(d,2)=1\\ d\equiv 0 \bmod M}} r(d)\left(\frac{d}{n}\right) F\left(\frac{d}{Z}\right) = \sum_{d\equiv 0 \bmod M} r(d)\left(\frac{d}{n}\right) F\left(\frac{d}{Z}\right) - \left(\frac{2}{n}\right)\sum_{d\equiv 0 \bmod M} r(d)\left(\frac{d}{n}\right) F\left(\frac{2d}{Z}\right), \tag{2}$$

since $r(2d)=r(d)$. Next, we focus on the sum

$$\sum_{d\equiv 0 \bmod M} r(d)\left(\frac{d}{n}\right) F\left(\frac{ad}{Z}\right),$$

for $a=1$ or 2. We take out the real character by imposing congruence conditions on a_1 and a_2 where $a_1^2+a_2^2=d$. This gives us, since $(M,2n)=1$, that the latter sum equals

$$\sum_{\substack{b_1,b_2 \bmod Mn\\ b_1^2+b_2^2\equiv 0 \bmod M}}\left(\frac{b_1^2+b_2^2}{n}\right)\sum_{\substack{a_1\equiv b_1 \bmod Mn\\ a_2\equiv b_2 \bmod Mn}} F\left(\frac{(a_1^2+a_2^2)a}{Z}\right).$$

The inner sum over a_1, a_2 by the Voronoi summation formula gives (by writing $a = b_1 + Mn\ell_1$ and $b = b_2 + Mn\ell_2$)

$$\sum_{\substack{a_1\equiv b_1 \bmod Mn\\ a_2\equiv b_2 \bmod Mn}} F\left(\frac{(a_1^2+a_2^2)a}{Z}\right) = \frac{Z}{2aM^2n^2}\sum_{(k_1,k_2)\in\mathbb{Z}^2}\check{F}\left(\frac{(k_1^2+k_2^2)Z}{aM^2n^2}\right) e\left(\frac{k_1b_1}{Mn}+\frac{k_2b_2}{Mn}\right),$$

where $\check{F}(u)$ is the Bessel transform of F given in (1). Now summing over $b_1, b_2 \bmod Mn$, we get that

$$\sum_{d\equiv 0 \bmod M} r(d)\left(\frac{d}{n}\right) F\left(\frac{ad}{Z}\right) = \frac{Z}{2aM^2n^2}\sum_{(k_1,k_2)\in\mathbb{Z}^2} G_{(k_1,k_2)}(M,n)\check{F}\left(\frac{(k_1^2+k_2^2)Z}{aM^2n^2}\right),$$

where $G_{(k_1,k_2)}(M,n)$ is as in the statement of the result. Putting this in (2), we write $k_1 = \frac{k_1'+k_2'}{2}$, and $k_2 = \frac{k_1'-k_2'}{2}$ for $k_1' \equiv k_2' \bmod 2$, and use the following relation that we prove below

$$G_{\left(\frac{k_1'+k_2'}{2},\frac{k_1'-k_2'}{2}\right)}(M,n) = \left(\frac{2}{n}\right) G_{(k_1',k_2')}(M,n),$$

to get the result.

To prove $G_{\left(\frac{k_1+k_2}{2},\frac{k_1-k_2}{2}\right)}(M,n) = \left(\frac{2}{n}\right) G_{(k_1,k_2)}(M,n)$ for $k_1 \equiv k_2 \bmod 2$, we note that

$$\left(\frac{2}{n}\right) G_{(k_1,k_2)}(M,n) = \sum_{\substack{b_1,b_2 \bmod Mn \\ b_1^2+b_2^2 \equiv 0 \bmod M}} \left(\frac{2(b_1^2+b_2^2)}{n}\right) e\left(\frac{k_1 b_1}{Mn} + \frac{k_2 b_2}{Mn}\right).$$

We can then do a change of variables $b_1 \to b_1 + b_2$ and $b_2 \to b_1 - b_2$ which yields

$$\left(\frac{2}{n}\right) G_{(k_1,k_2)}(M,n) = \sum_{\substack{b_1,b_2 \bmod Mn \\ b_1^2+b_2^2 \equiv 0 \bmod M}} \left(\frac{b_1^2+b_2^2}{n}\right) e\left(\frac{\bar{2}(k_1+k_2)b_1}{Mn} + \frac{\bar{2}(k_1-k_2)b_2}{Mn}\right),$$

where $\bar{2}$ denotes the inverse of $2 \bmod Mn$. Next, by reciprocity we have

$$\frac{\bar{2}}{Mn} \equiv \frac{1}{2Mn} - \frac{1}{2} \bmod 1.$$

Plugging this in, we get

$$\begin{aligned}\left(\frac{2}{n}\right) G_{(k_1,k_2)}(M,n) &= \sum_{\substack{b_1,b_2 \bmod Mn \\ b_1^2+b_2^2 \equiv 0 \bmod M}} \left(\frac{b_1^2+b_2^2}{n}\right) e\left(\frac{(k_1+k_2)b_1}{2Mn} + \frac{(k_1-k_2)b_2}{2Mn}\right) \\ &= G_{\left(\frac{k_1+k_2}{2},\frac{k_1-k_2}{2}\right)}(M,n),\end{aligned}$$

since $k_1 \equiv k_2 \bmod 2$, and so the term coming from the $-\frac{1}{2}$ factor of the reciprocity law disappears. □

Remark 9. *Following a similar argument for any primitive positive-definite binary quadratic form $Q(x,y) = ax^2+bxy+cy^2$ with discriminant Δ, one obtains a general Voronoi summation formula of the form*

$$\sum_{d \equiv h \bmod M} r_Q(d) \left(\frac{d}{n}\right) F\left(\frac{d}{Z}\right) = \frac{Z}{\sqrt{|\Delta|}M^2n^2} \sum_{(k_1,k_2)\in\mathbb{Z}^2} G_{(k_1,k_2)}(M,n,h;Q) \check{F}\left(\frac{4Z \cdot Q^*(k_1,k_2)}{|\Delta|M^2n^2}\right),$$

where $Q^*(x,y) = cx^2 - bxy + ay^2$ is the dual quadratic form, $\check{F}(x)$ is the Bessel transform defined in (1), and

$$G_{(k_1,k_2)}(M,n,h;Q) := \sum_{\substack{b_1,b_2 \bmod Mn \\ Q(b_1,b_2) \equiv h \bmod M}} \left(\frac{Q(b_1,b_2)}{n}\right) e\left(\frac{k_1 b_1 + k_2 b_2}{Mn}\right),$$

are the two-dimensional Gauss sums attached to $Q(x,y)$.

The next lemma shows a convenient splitting of the two-dimensional Gauss sums we encountered in Proposition 8. In particular, it turns out that one may separate the local conditions appropriately.

Lemma 10. *Let $k_1, k_2 \in \mathbb{Z}$, and n coprime to M. Then we have that*

$$G_{(k_1,k_2)}(M,n) = G_{(k_1,k_2)}(n) \cdot \rho_{(k_1,k_2)}(M),$$

where

$$G_{(k_1,k_2)}(n) := \sum_{b_1,b_2 \bmod n} \left(\frac{b_1^2+b_2^2}{n}\right) e\left(\frac{k_1 b_1}{n} + \frac{k_2 b_2}{n}\right),$$

and

$$\rho_{(k_1,k_2)}(M) := \sum_{\substack{b_1,b_2 \bmod M \\ b_1^2+b_2^2 \equiv 0 \bmod M}} e\left(\frac{k_1 b_1}{M} + \frac{k_2 b_2}{M}\right).$$

Proof. We start by noting that if $(b_1, b_2) \in \mathbb{Z}/Mn\mathbb{Z}$ is such that $b_1^2 + b_2^2 \equiv 0 \bmod M$, then by the Chinese remainder theorem, one can write $b_1 = b_1' M + b_1'' n$ for $b_1' \bmod n$ and $b_1'' \bmod M$ since $\gcd(n, M) = 1$, and similarly for b_2. Using this, we get

$$\begin{aligned}
&G_{(k_1,k_2)}(M,n) \\
&= \sum_{b_1',b_2' \bmod n} \sum_{\substack{b_1'',b_2'' \bmod M \\ b_1''^2+b_2''^2 \equiv 0 \bmod M}} \left(\frac{(b_1'M+b_1'')^2+(b_2'M+b_2'')^2}{n}\right) e\left(\frac{k_1(b_1'M+b_1'')}{Mn} + \frac{k_2(b_2'M+b_2'')}{Mn}\right) \\
&= \sum_{b_1',b_2' \bmod n} \left(\frac{b_1'^2+b_2'^2}{n}\right) e\left(\frac{k_1 b_1'}{n} + \frac{k_2 b_2'}{n}\right) \cdot \sum_{\substack{b_1'',b_2'' \bmod M \\ b_1''^2+b_2''^2 \equiv 0 \bmod M}} e\left(\frac{k_1 b_1''}{M} + \frac{k_2 b_2''}{M}\right),
\end{aligned}$$

which gives us the result. □

Denote by

$$\rho(n) := \rho_{(0,0)}(n) = \#\{(u_1,u_2) \bmod n : u_1^2 + u_2^2 \equiv 0 \bmod n\}. \tag{3}$$

We now prove multiplicative properties of our Gauss sums and how they behave locally.

Proposition 11. *If m and n are relatively prime odd integers, then*

$$G_{(k_1,k_2)}(mn) = G_{(k_1,k_2)}(m) G_{(k_1,k_2)}(n).$$

Moreover, if p^α is the highest power of p dividing $\gcd(k_1,k_2)$ (setting $\alpha = \infty$ if $(k_1,k_2) = (0,0)$), then we have for $\beta \geq 1$,

$$G_{(k_1,k_2)}(p^\beta) = \begin{cases} 0 & \text{if } \beta \leq \alpha \text{ is odd} \\ p^{2(\beta-1)}(p^2 - \rho(p)) & \text{if } \beta \leq \alpha \text{ is even} \\ p^{2\alpha+1}\left(\frac{(k_1^2+k_2^2)p^{-2\alpha}}{p}\right) & \text{if } \beta = \alpha+1 \text{ is odd} \\ -p^{2\alpha}\left(\frac{-1}{p}\right)(p \cdot \delta_p((k_1^2+k_2^2)p^{-2\alpha}) - 1) & \text{if } \beta = \alpha+1 \text{ is even} \\ 0 & \text{if } \beta \geq \alpha+2, \end{cases}$$

where $\delta_p(n)$ is 1 if $n \equiv 0 \bmod p$ and 0 otherwise, and $\rho(n)$ is as in (3).

Proof. We first prove the multiplicativity. Since $\gcd(m,n)=1$, write $b_1 = mb_1' + nb_1''$ for $b_1' \bmod n$ and $b_1'' \bmod m$ by the Chinese remainder theorem. And similarly for $b_2 = mb_2' + nb_2''$. We then have

$$G_{(k_1,k_2)}(mn) = \sum_{\substack{b_1',b_2' \bmod n \\ b_1'',b_2'' \bmod m}} \left(\frac{(b_1'm+b_1''n)^2+(b_2'm+b_2''n)^2}{mn}\right) e\left(\frac{k_1(b_1'm+b_1''n)+k_2(b_2'm+b_2''n)}{mn}\right).$$

Using the multiplicativity of the Kronecker symbol and the linearity in the exponential phase, we get that this is $G_{(k_1,k_2)}(m)G_{(k_1,k_2)}(n)$ as desired.

We now prove the second part of the lemma. If $\beta = \alpha+1$, then

$$\begin{aligned}\sum_{b_1,b_2 \bmod p^\beta} &\left(\frac{b_1^2+b_2^2}{p^\beta}\right) e\left(\frac{k_1b_1}{p^\beta}+\frac{k_2b_2}{p^\beta}\right) \\ &= \sum_{\ell_1,\ell_2 \bmod p}\left(\frac{\ell_1^2+\ell_2^2}{p^\beta}\right)\sum_{a_1,a_2 \bmod p^{\beta-1}} e\left(\frac{k_1(a_1p+\ell_1)+k_2(a_2p+\ell_2)}{p^\beta}\right) \\ &= p^{2(\beta-1)}\sum_{\ell_1,\ell_2 \bmod p}\left(\frac{\ell_1^2+\ell_2^2}{p^\beta}\right) e\left(\frac{k_1\ell_1+k_2\ell_2}{p^\beta}\right).\end{aligned}$$

If β is even, then the sum above is

$$\sum_{\substack{\ell_1,\ell_2 \bmod p \\ \ell_1^2+\ell_2^2 \not\equiv 0 \bmod p}} e\left(\frac{k_1\ell_1+k_2\ell_2}{p^\beta}\right) = -\sum_{\substack{\ell_1,\ell_2 \bmod p \\ \ell_1^2+\ell_2^2 \equiv 0 \bmod p}} e\left(\frac{k_1\ell_1+k_2\ell_2}{p^\beta}\right),$$

since the full sum is 0. We write $k_1' = k_1/p^\alpha$ and $k_2' = k_2/p^\alpha$. By orthogonality, we get

$$\sum_{\substack{\ell_1,\ell_2 \bmod p \\ \ell_1^2+\ell_2^2 \equiv 0 \bmod p}} e\left(\frac{k_1\ell_1+k_2\ell_2}{p^\beta}\right) = \frac{1}{p}\sum_{a \bmod p}\sum_{\ell_1,\ell_2 \bmod p} e\left(\frac{a(\ell_1^2+\ell_2^2)+k_1'\ell_1+k_2'\ell_2}{p}\right).$$

This equals

$$\frac{1}{p}\sum_{a \bmod p}\left(\sum_{\ell_1 \bmod p} e\left(\frac{a\ell_1^2+k_1'\ell_1}{p}\right)\right)\left(\sum_{\ell_2 \bmod p} e\left(\frac{a\ell_2^2+k_2'\ell_2}{p}\right)\right).$$

Note that the term $a=0$ gives 0. By the theory of Gauss sums, for $a \neq 0$, these sums over ℓ_1, ℓ_2 evaluate to

$$\left(\frac{a}{p}\right)\tau_p e\left(\frac{-\overline{4a}k_i'^2}{p}\right),$$

where τ_p is the usual Gauss sum and so $\tau_p^2 = \left(\frac{-1}{p}\right)p$. Hence, we get the fourth case of the lemma, since we are left with

$$-p^{2(\beta-1)}\left(\frac{-1}{p}\right)\sum_{\substack{a \bmod p \\ a\neq 0}} e\left(\frac{-\overline{4a}(k_1'^2+k_2'^2)}{p}\right),$$

and this sum is the Ramanujan sum.

If β is odd, we have

$$p^{2(\beta-1)} \sum_{\ell_1,\ell_2 \bmod p} \left(\frac{\ell_1^2+\ell_2^2}{p}\right) e\left(\frac{k_1\ell_1+k_2\ell_2}{p^\beta}\right).$$

Similar to the even case, we can write the sum as

$$\sum_{\ell_1,\ell_2 \bmod p} \left(\frac{\ell_1^2+\ell_2^2}{p}\right) e\left(\frac{k_1'\ell_1+k_2'\ell_2}{p}\right).$$

We rewrite the Legendre symbol as

$$\left(\frac{\ell_1^2+\ell_2^2}{p}\right) = \frac{1}{\tau_p} \sum_{\substack{a \bmod p \\ a\neq 0}} \left(\frac{a}{p}\right) e\left(\frac{a(\ell_1^2+\ell_2^2)}{p}\right).$$

Putting this in, similar to the even case, we get sums of the form

$$\sum_{\ell_i \bmod p} e\left(\frac{a\ell_i^2+k_i'\ell_i}{p}\right) = e\left(-\frac{\overline{4a}k_1^2}{p}\right) \sum_{\ell_1} e\left(\frac{a(\ell_i+\overline{2a}k_i')^2}{p}\right).$$

Now changing the variable of summation $\ell_i + k_i'\overline{a} \to \ell_i$, the inner sum evaluates to $\left(\frac{a}{p}\right)\tau_p$ by the classical theory of Gauss sums. Putting everything back and using $\tau_p^2 = \left(\frac{-1}{p}\right)p$ again, yields

$$\sum_{\ell_1,\ell_2 \bmod p} \left(\frac{\ell_1^2+\ell_2^2}{p}\right) e\left(\frac{k_1'\ell_1+k_2'\ell_2}{p}\right) = \frac{\left(\frac{-1}{p}\right)p}{\tau_p} \sum_{\substack{a \bmod p \\ a\neq 0}} \left(\frac{a}{p}\right) e\left(\frac{-\overline{4a}(k_1'^2+k_2'^2)}{p}\right).$$

Lastly, we do the change of variables $-\overline{4a} \to u$ which gives $\left(\frac{a}{p}\right) = \left(\frac{-\overline{4u}}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{u}{p}\right)$. Hence, we obtain

$$\frac{p}{\tau_p} \sum_{\substack{u \bmod p \\ u\neq 0}} \left(\frac{u}{p}\right) e\left(\frac{u(k_1'^2+k_2'^2)}{p}\right).$$

Evaluating this Gauss sum gives the third case of the lemma.

The other cases are done in a similar manner. □

Definition 12. *We say a function $f(x,y)$ over $\mathbb{Z}^2$ is jointly multiplicative if $f(m_1n_1, m_2n_2) = f(m_1,m_2)f(n_1,n_2)$ whenever $\gcd(m_1m_2, n_1n_2) = 1$.*

The following observation will be found useful.

Lemma 13. *Let ℓ, n be odd integers and $(k_1,k_2) \in \mathbb{Z}^2 \setminus (0,0)$. Then $G_{(\ell k_1, \ell k_2)}(n)$ is jointly multiplicative in ℓ and n.*

Proof. Define $f(n,\ell) = G_{(\ell k_1,\ell k_2)}(n)$. We want to show that this is jointly multiplicative. Given $\gcd(n_1\ell_1, n_2\ell_2) = 1$, note that we have

$$f(n_1n_2, \ell_1\ell_2) = \sum_{b_1,b_2 \bmod n_1n_2} \left(\frac{b_1^2+b_2^2}{n_1n_2}\right) e\left(\frac{\ell_1\ell_2k_1b_1+\ell_1\ell_2k_2b_2}{n_1n_2}\right).$$

Next, by the Chinese Remainder Theorem, we can write $b_1 = b_1' n_1 + b_1'' n_2$ for $b_1' \bmod n_2$ and $b_1'' \bmod n_1$ since $\gcd(n_1, n_2) = 1$. And likewise for $b_2 \bmod n_2$. Hence, similar to the first part of Proposition 11, one gets

$$f(n_1 n_2, \ell_1 \ell_2) = \sum_{\substack{b_1', b_2' \bmod n_2 \\ b_1' b_2'' \bmod n_1}} \left(\frac{b_1'^2 + b_2'^2}{n_2}\right) \left(\frac{b_1''^2 + b_2''^2}{n_1}\right) e\left(\frac{\ell_1 \ell_2 b_1'' + \ell_1 \ell_2 b_2''}{n_1} + \frac{\ell_1 \ell_2 b_1' + \ell_1 \ell_2 b_2'}{n_2}\right).$$

We notice that since $\gcd(\ell_1, n_2) = 1$ and $\gcd(\ell_2, n_1) = 1$, we can do change of variables $\ell_1 b_1' \to b_1'$, $\ell_1 b_2' \to b_2'$ and $\ell_2 b_1'' \to b_1''$, $\ell_2 b_2'' \to b_2''$. Then using

$$\left(\frac{b_1^2 + b_2^2}{n}\right) = \left(\frac{(\bar{\ell} b_1')^2 + (\bar{\ell} b_2')^2}{n}\right) = \left(\frac{b_1'^2 + b_2'^2}{n}\right),$$

we get that this gives $f(n_1, \ell_1) f(n_2, \ell_2)$. □

Lemma 14. *Let $k_1, k_2 \in \mathbb{Z}$. If m and n are relatively prime odd integers, then we have $\rho_{(k_1,k_2)}(mn) = \rho_{(k_1,k_2)}(m)\rho_{(k_1,k_2)}(n)$. Further, for any ℓ coprime to n, we have $\rho_{(\ell k_1, \ell k_2)}(n) = \rho_{(k_1,k_2)}(n)$. Moreover, we have the bound $|\rho_{(k_1,k_2)}(n)| \le \rho(n) \le \tau(n) n$, where $\rho(n)$ is defined in (3).*

Proof. The multiplicativity comes from the Chinese Remainder Theorem as in the proof of the previous lemma.

For the second property, write

$$\rho_{(\ell k_1, \ell k_2)}(n) = \sum_{\substack{b_1, b_2 \bmod n \\ b_1^2 + b_2^2 \equiv 0 \bmod n}} e\left(\frac{\ell k_1 b_1 + \ell k_2 b_2}{n}\right).$$

By a change of variable $\ell b_1 \to b_1$ and $\ell b_2 \to b_2$, we get the same summand without the ℓ factors, but with the additional condition that $\ell^2(b_1^2 + b_2^2) \equiv 0 \bmod n$. Since $\gcd(\ell, n) = 1$, this will imply that $\rho_{(\ell k_1, \ell k_2)}(n) = \rho_{(k_1,k_2)}(n)$.

For the last bounds, note that $|\rho_{(k_1,k_2)}(n)| \le \rho(n)$ is the trivial bound. For the second bound, by multiplicativity, we just need to study the prime powers. Note that $\rho(p^\alpha) = \{(u, v) \in (\mathbb{Z}/p^\alpha\mathbb{Z})^2 : u^2 + v^2 \equiv 0 \bmod p^\alpha\}$. Hence, for each $u \in \mathbb{Z}/p^\alpha\mathbb{Z}$, we have at most 2 choices for v, and so $\rho(p^\alpha) \le 2p^\alpha$. The result then follows. □

4. Proof of Theorem 1

We want to compute

$$\sum_{\substack{dSF \\ (d,2N)=1 \\ w(f \otimes \chi_{8d}) = -1}} r(d) L'(1/2, f \otimes \chi_{8d}) \psi\left(\frac{8d}{X}\right).$$

This equals

$$\frac{1}{2} \sum_{\substack{dSF \\ (d,2N)=1}} r(d) L'(1/2 f \otimes \chi_{8d}) \psi\left(\frac{8d}{X}\right) - \frac{w(f)}{2} \sum_{\substack{dSF \\ (d,2N)=1}} r(d) \chi_{8d}(-N) L'(1/2, f \otimes \chi_{8d}) \psi\left(\frac{8d}{X}\right).$$

Opening up $L'(1/2, f \otimes \chi_{8d})$ using Lemma 5, and truncating both sums to $n \le U = \frac{X}{(\log X)^A}$, for some large $A > 0$, we will get

$$L'(1/2, f \otimes \chi_{8d}) = \mathcal{A}_U(f \otimes \chi_{8d}) + \mathcal{B}_U(f \otimes \chi_{8d}),$$

where

$$\mathcal{A}_U(f\otimes\chi_{8d})=(1-i^{\kappa}\eta\chi_{8d}(N))\sum_{n\geq 1}\frac{\lambda_f(n)\chi_{8d}(n)}{\sqrt{n}}W\left(\frac{n}{U}\right),$$

and

$$\mathcal{B}_U(f\otimes\chi_{8d})=(1-i^{\kappa}\eta\chi_{8d}(N))\sum_{n\geq 1}\frac{\lambda_f(n)\chi_{8d}(n)}{\sqrt{n}}\left(W\left(\frac{n}{8d}\right)-W\left(\frac{n}{U}\right)\right).$$

4.1 THE $\mathcal{A}_U$ SUM

Let $N'=1$ or $N'=N$. We want to study

$$T(N',h):=\sum_{\substack{d\ SF\\(d,2N)=1}}r(d)\sum_{n\geq 1}\frac{\lambda_f(n)\chi_{8d}(nN')}{\sqrt{n}}h(d,n)$$

where

$$h(d,n):=W\left(\frac{n}{U}\right)\psi\left(\frac{8d}{X}\right).$$

We detect the SF and coprimality conditions to get

$$\begin{aligned}T(N',h)&=\sum_{(a_1,2NN')=1}\mu(a_1)\sum_{\substack{a_2|N\\(a_2,N')=1}}\mu(a_2)\sum_{(d,2)=1}r(da_1^2a_2)\sum_{(n,a_1)=1}\frac{\lambda_f(n)\chi_{8da_2}(nN')}{\sqrt{n}}h(da_1^2a_2,n)\\&=T_1(N',h)+T_2(N',h),\end{aligned}$$

where

$$T_1(N',h):=\sum_{\substack{(a_1,2NN')=1\\a_1\leq Y}}\mu(a_1)\sum_{\substack{a_2|N\\(a_2,N')=1}}\mu(a_2)\sum_{(d,2)=1}r(da_1^2a_2)\sum_{(n,a_1)=1}\frac{\lambda_f(n)\chi_{8da_2}(nN')}{\sqrt{n}}h(da_1^2a_2,n),$$

and

$$T_2(N',h):=\sum_{\substack{(a_1,2NN')=1\\a_1>Y}}\mu(a_1)\sum_{\substack{a_2|N\\(a_2,N')=1}}\mu(a_2)\sum_{(d,2)=1}r(da_1^2a_2)\sum_{(n,a_1)=1}\frac{\lambda_f(n)\chi_{8da_2}(nN')}{\sqrt{n}}h(da_1^2a_2,n),$$

for $Y=(\log X)^{A/10}$.

The main term will come from the T_1 sum, while the T_2 sum will be an error.

4.1.1 Bounding $T_2(N',h)$

We start by adding the coprimality condition $a_2|N$ again, and then writing $d=d_1d_2^2$ for d_1 SF and $d_2\in\mathbb{Z}_{>0}$. After exchanging sums and using the fact that $r(d_1d_2^2a_1^2)<\tau(d_1)\tau(d_2^2)\tau(a_1^2)$, we

use Cauchy-Schwarz to get

$$T_2(N',h) \ll \sum_{\substack{(a_1,2NN')=1\\ a_1>Y}} \mu(a_1)\tau(a_1^2) \sum_{\substack{(d_2,2N)=1\\ d_2\leq\sqrt{X}}} \tau(d_2^2) \left(\sum_{\substack{(d_1,2N)=1\\ d_1SF\\ d_1\leq X/(a_1d_2)^2}} \tau^2(d_1) \right)^{1/2} \times \left(\sum_{\substack{(d_1,2N)=1\\ d_1SF\\ d_1\leq X/(a_1d_2)^2}} \left| \sum_{(n,d_2)=1} \frac{\lambda_f(n)\chi_{8d_1}(nN')}{\sqrt{n}} W\left(\frac{n}{U}\right) \right|^2 \right)^{1/2}.$$

Now, employing [13, Proposition 3.2] (and in particular the bound in Section 5.5 in their paper) together with the fact that

$$\sum_{n\leq x} \tau^2(n) \asymp x\log^3 x,$$

we get

$$T_2(N',h) \ll \frac{X\log^3 X}{Y^{1-\epsilon}}.$$

4.1.2 Analyzing $T_1(N',h)$

We start by changing the order of summations in $T_1(N',h)$. Using $\chi_{8d}(nN') = \left(\frac{8d}{nN'}\right)$, one has

$$T_1(N',h) = \sum_{\substack{(a_1,2N)=1\\ a_1\leq Y}} \mu(a_1) \sum_{\substack{a_2|N\\ (a_2,N')=1}} \mu(a_2) \sum_{(n,2a_1a_2)=1} \frac{\lambda_f(n)}{\sqrt{n}} \sum_{(d,2)=1} r(da_1^2a_2) \left(\frac{8da_1^2a_2}{nN'}\right) h(da_1^2a_2,n).$$

Next, we use Proposition 8 to get that the inner sum is

$$\frac{X}{4a_1^4a_2^2n^2N'^2} \sum_{(k_1,k_2)\in\mathbb{Z}^2} (-1)^{k_1+k_2} G_{(k_1,k_2)}(a_1^2a_2,nN') \check{h}\left(\frac{(k_1^2+k_2^2)X}{2a_1^4a_2^2n^2N'^2},n\right),$$

where

$$\check{h}\left(\frac{(k_1^2+k_2^2)X}{2a_1^4a_2^2n^2N'^2},n\right) = 2\pi\int_0^\infty h(Xt,n) J_0\left(2\pi\sqrt{\frac{(k_1^2+k_2^2)Xt}{2a_1^4a_2^2n^2N'^2}}\right) dt,$$

is the Bessel transform of h. We can further use Lemma 10 for splitting the function G. Here, the $(k_1,k_2)=(0,0)$ term will give us the main term, and the rest will be an error.

Main Term $(k_1,k_2)=(0,0)$: Denote by $T_{10}(N',h)$ the $(k_1,k_2)=(0,0)$ term. That is,

$$T_{10}(N',h) = \frac{X}{4N'^2} \sum_{\substack{(a_1,2NN')=1\\ a_1\leq Y}} \frac{\mu(a_1)\rho(a_1^2)}{a_1^4} \sum_{\substack{a_2|N\\ (a_2,N')=1}} \frac{\mu(a_2)\rho(a_2)}{a_2^2} \sum_{(n,2a_1a_2)=1} \frac{\lambda_f(n)\cdot G(nN')}{n^{5/2}} h_1(n),$$

where recall

$$G(nN') = \sum_{b_1,b_2 \bmod nN'} \left(\frac{b_1^2+b_2^2}{nN'}\right),$$

and

$$\rho(m) = \#\{(u_1, u_2) \bmod m : u_1^2 + u_2^2 \equiv 0 \bmod m\}$$

are multiplicative functions, and

$$h_1(n) := \int_0^\infty h(Xt, n)dt.$$

Next, we can write this in a cleaner form, since $G(n) \leq n^2$,

$$T_{10}(N', h) = CX \sum_{(n,2)=1} \frac{\lambda_f(n)}{\sqrt{n}} \frac{G(nN')}{n^2 N'^2} \prod_{p|nNN'} \left(1 - \frac{\rho(p^2)}{p^4}\right)^{-1} \prod_{\substack{p|N \\ p\nmid nN'}} \left(1 - \frac{\rho(p)}{p^2}\right) \cdot h_1(n) + O\left(\frac{X}{Y} \sum_{n=\square} \frac{\tau(n)}{n^{1/2}} |h_1(n)|\right),$$

where

$$C := \frac{1}{4} \prod_{p>2} \left(1 - \frac{\rho(p^2)}{p^4}\right).$$

To see how large is the big-O term, we use bounds on $\psi(x)$ and Lemma 6 to get the following bound on the partial derivatives of $h(\cdot, \cdot)$ that follow from integration by parts,

$$h^{(i,j)}(x, y) \ll_{j,B} \frac{1}{x^i y^j} \left(\log \frac{\kappa NU}{n}\right) \left(1 + \frac{n}{\kappa\sqrt{N}U}\right)^{-B} \left(1 + \frac{d}{X}\right)^{-B}, \tag{4}$$

for any $B > 0$. Using this in $h_1(n)$, we see that the big-O term is at most $\ll \frac{X}{Y}(\log X)^4$.

Using Proposition 11 and noting that $N' \in \{1, N\}$, we get

$$T_{10}(N', h) = CX \sum_{\substack{(n,2)=1 \\ nN'=\square}} \frac{\lambda_f(n)}{n^{1/2}} \prod_{p|nNN'} \left(1 - \frac{\rho(p^2)}{p^4}\right)^{-1} \prod_{p|nNN'} \left(1 - \frac{\rho(p)}{p^2}\right) h_1(n) + O\left(\frac{X}{(\log X)^{A/100}}\right).$$

We recall that $h(x, y) = W(y/U)\psi(8x/X)$, hence this gives us

$$T_{10}(N', h) = \frac{C}{16}\hat{\psi}(0)X \sum_{\substack{(n,2)=1 \\ nN'=\square}} \frac{\lambda_f(n)}{n^{1/2}} \prod_{p|nNN'} \left(1 - \frac{\rho(p)}{p^2}\right) \left(1 - \frac{\rho(p^2)}{p^4}\right)^{-1} W\left(\frac{n}{U}\right) + O\left(\frac{X}{(\log X)^{A/100}}\right),$$

where

$$\hat{\psi}(0) = 16 \int_0^\infty \psi(8t)\, dt$$

is the Fourier transform of ψ at 0. By the definition of $W(\cdot)$,

$$T_{10}(N', h) = \frac{C}{16}\hat{\psi}(0)X \cdot \frac{1}{2i\pi} \int_{(3)} \frac{\Gamma(u + \kappa/2)}{\Gamma(\kappa/2)} \left(\frac{2\pi}{U\sqrt{N}}\right)^{-u} \frac{\mathcal{Z}(u)}{u^2} du + O\left(\frac{X}{(\log X)^{A/100}}\right),$$

where

$$\mathcal{Z}(u) := \sum_{\substack{(n,2)=1\\ nN'=\square}} \frac{\lambda_f(n)}{n^{1/2+u}} \prod_{p|nNN'} \left(1-\frac{\rho(p)}{p^2}\right)\left(1-\frac{\rho(p^2)}{p^4}\right)^{-1}.$$

The Dirichlet series $\mathcal{Z}(u)$ has the following Euler product

$$\begin{aligned}\mathcal{Z}(u) = &\prod_{\substack{p\nmid 2NN'\\ p\equiv 1 \bmod 4}} \left(1+\frac{p(p-1)}{p^2-p+1}\left[\frac{1}{2}\left(1-\frac{\lambda_f(p)}{p^{1/2+u}}+\frac{1}{p^{1+2u}}\right)^{-1}+\frac{1}{2}\left(1+\frac{\lambda_f(p)}{p^{1/2+u}}+\frac{1}{p^{1+2u}}\right)^{-1}-1\right]\right)\\ &\prod_{\substack{p\nmid 2NN'\\ p\equiv 3 \bmod 4}} \left(1+\frac{p(p+1)}{p^2+p+1}\left[\frac{1}{2}\left(1-\frac{\lambda_f(p)}{p^{1/2+u}}+\frac{1}{p^{1+2u}}\right)^{-1}+\frac{1}{2}\left(1+\frac{\lambda_f(p)}{p^{1/2+u}}+\frac{1}{p^{1+2u}}\right)^{-1}-1\right]\right)\\ &\prod_{p|NN'} \frac{p(p^2-\rho(p))}{p^3-\rho(p)}\left[\frac{1}{2}\left(1-\frac{\lambda_f(p)}{p^{1/2+u}}\right)^{-1}+(-1)^{\mathrm{ord}_p(N')}\frac{1}{2}\left(1+\frac{\lambda_f(p)}{p^{1/2+u}}\right)^{-1}\right],\end{aligned}$$

which comes from Hecke recurrences for $\lambda_f(n)$ and the facts that $\rho(p) = (1+\chi_4(p))(p-1)+1$, and $\rho(p^2) = p\cdot\rho(p)$. We can write this in the form

$$\mathcal{Z}(u) = L(1+2u, \mathrm{sym}^2 f)\cdot A(u,N,N'), \tag{5}$$

where

$$L(s,\mathrm{sym}^2 f) = \zeta(2s)\sum_{n=1}^{\infty}\frac{\lambda_f(n^2)}{n^s}, \tag{6}$$

is the symmetric square L-function for f, and

$$A(u,N,N') = \prod_{p\nmid 2NN'}\left(1+O\left(\frac{1}{p^{2+2\,\mathrm{Re}(u)}}+\frac{1}{p^{2+4\,\mathrm{Re}(u)}}\right)\right)\cdot\prod_{p|2NN'}\left(1+O\left(\frac{1}{p^{1+2\,\mathrm{Re}(u)}}+\frac{1}{p}\right)\right)$$

is an absolutely convergent product for $\mathrm{Re}(u) > -1/4$. Note that $A(u,N,N')$ only depends on $\mathrm{ord}_p(N')$ for $p|NN'$. We have also used that for $\mathrm{Re}(s)>1$,

$$L(s,\mathrm{sym}^2 f) = \prod_p \left(1-\frac{\alpha_f(p)^2}{p^s}\right)^{-1}\left(1-\frac{\alpha_f(p)\beta_f(p)}{p^s}\right)^{-1}\left(1-\frac{\beta_f(p)^2}{p^s}\right)^{-1},$$

where $\alpha_f(p)+\beta_f(p) = \lambda_f(p)$ and $\alpha_f(p)\beta_f(p) = 1$.

Next, we move the contour in the integral of $T_{10}(N',h)$ to $\mathrm{Re}(u) = -\frac{1}{6}$, and pick up the double pole at $u=0$. This gives us

$$\begin{aligned}T_{10}(N',h) = \frac{C}{16}X\cdot\hat{\psi}(0)L(1,\mathrm{sym}^2 f)A(0,N,N')\left(\log\frac{U\sqrt{N}}{2\pi}+\frac{A'(0,N,N')}{A(0,N,N')}+2\frac{L'(1,\mathrm{sym}^2 f)}{L(1,\mathrm{sym}^2 f)}\right)\\ +\frac{C}{16}X\cdot\hat{\psi}(0)\frac{1}{2i\pi}\int_{(-1/6)}\Gamma(u+1)\left(\frac{2\pi}{U\sqrt{N}}\right)^{-u}\frac{\mathcal{Z}(u)}{u^2}du.\end{aligned}$$

We use the convexity bound for the GL(3) object (see [8, Eqn (5.20)])

$$L(2/3+it,\mathrm{sym}^2 f) \ll (1+|t|)^{1+\varepsilon},$$

to bound the second term. This will give us an error $\ll_\varepsilon X^{5/6+\varepsilon}$.

4.1.3 Off-diagonal terms $(k_1,k_2)\neq(0,0)$

In this section, we study the following sum

$$T_{11}(N',h):=\frac{X}{4}\sum_{(k_1,k_2)\neq(0,0)}(-1)^{k_1+k_2}\sum_{\substack{(a_1,2N)=1\\ a_1\leq Y}}\frac{\mu(a_1)\rho_{(k_1,k_2)}(a_1^2)}{a_1^4}\sum_{a_2|N}\frac{\mu(a_2)\rho_{(k_1,k_2)}(a_2)}{a_2^2}$$
$$\times\sum_{(n,2a_1a_2)=1}\frac{\lambda_f(n)}{\sqrt{n}}\frac{G_{(k_1,k_2)}(nN')}{n^2N'^2}\check{h}\left(\frac{(k_1^2+k_2^2)X}{2a_1^4a_2^2n^2N'^2},n\right),$$

where recall

$$\rho_{(k_1,k_2)}(m)=\sum_{\substack{b_1,b_2 \bmod m\\ b_1^2+b_2^2\equiv 0 \bmod m}}e\left(\frac{k_1b_1}{m}+\frac{k_2b_2}{m}\right),$$

is a multiplicative function. We use Mellin inversion and then a change of variables to write

$$\begin{aligned}\check{h}(x,y)&=2\pi\int_0^\infty h(Xt,y)J_0(2\pi\sqrt{tx})dt\\&=\frac{1}{iX}\int_0^\infty J_0(2\pi\sqrt{xtX^{-1}})\int_{(-1/4)}\tilde{h}(1+s,y)t^{-s}ds\frac{dt}{t}\\&=\frac{1}{iX}\int_0^\infty J_0(t)\int_{(1/2)}\tilde{h}(1-s/2,y)\left(\frac{t}{2\pi\sqrt{xX^{-1}}}\right)^s ds\frac{dt}{t}\\&=\frac{1}{iX}\int_{(1/2)}\tilde{h}(1-s/2,y)\left(\int_0^\infty J_0(t)t^s\frac{dt}{t}\right)(2\pi\sqrt{xX^{-1}})^{-s}ds\\&=\frac{X^{-1}}{2i}\int_{(1/2)}\tilde{h}(1-s/2,y)\frac{\Gamma(s/2)}{\Gamma(1-s/2)}(\pi\sqrt{xX^{-1}})^{-s}ds,\end{aligned}$$

where we used [4, (17.43.16)] for the last line. We can further perform one more Mellin inversion in the y variable to get

$$\check{h}(x,y)=-\frac{X^{-1}}{4\pi}\int_{(1+\epsilon)}\int_{(1/2+\varepsilon)}\tilde{\tilde{h}}(1-s/2,u)y^{-u}(\pi\sqrt{xX^{-1}})^{-s}\frac{\Gamma(s/2)}{\Gamma(1-s/2)}\,du\,ds,$$

where

$$\tilde{\tilde{h}}(s,u)=\int_{\mathbb{R}^2}h(t,v)t^sv^u\frac{dt}{t}\frac{dv}{v},$$

and it satisfies using integration by parts several times and the bounds in (4) on the partial derivatives of $h(\cdot,\cdot)$,

$$\tilde{\tilde{h}}(s,u)\ll\frac{X^{\mathrm{Re}(s)}(2U\sqrt{N})^{\mathrm{Re}(u)}}{|u|(1+|s|)^{98}(1+|u|)^{98}}. \tag{7}$$

Now, take the (k_1,k_2) sum of $T_{11}(N',h)$ inside, and split $k_1=\ell k_1'$ and $k_2=\ell k_2'$ where $\gcd(\ell,2a_1a_2)=1$, and further define

$$\mathcal{X}_{N'}(\alpha,\beta,a_1,a_2,k_1',k_2'):=\sum_{(\ell,2a_1a_2)=1}\sum_{(n,2a_1a_2)=1}\frac{\lambda_f(n)}{n^\alpha\ell^\beta}\cdot\frac{G_{(\ell k_1',\ell k_2')}(nN')}{n^2N'^2}.$$

Since $(\ell, a) = 1$ for $a \in \{a_1^2, a_2\}$, we have by Lemma 14 that $\rho_{(\ell k_1', \ell k_2')}(a) = \rho_{(k_1', k_2')}(a)$, and using a change of variable, we get

$$T_{11}(N', h) = -\frac{1}{16\pi} \sum_{\substack{(a_1, 2NN')=1 \\ a_1 \le Y}} \frac{\mu(a_1)}{a_1^4} \sum_{\substack{a_2 | N \\ (a_2, N')=1}} \frac{\mu(a_2)}{a_2^2} \sum_{\substack{(k_1', k_2') \neq (0,0) \\ p | \gcd(k_1', k_2') \implies p | 2a_1a_2}} (-1)^{k_1'+k_2'} \rho_{(k_1', k_2')}(a_1^2) \rho_{(k_1', k_2')}(a_2)$$

$$\times \int_{(1/2+\epsilon)} \int_{(1+\varepsilon)} \tilde{\tilde{h}}(1-s/2, u+s) \left(\frac{a_1^2 a_2 N'}{\sqrt{k_1'^2 + k_2'^2}} \right)^s \frac{\Gamma(s/2)}{\pi^s \cdot \Gamma(1-s/2)} \mathcal{X}_{N'}(1/2+u, s, a_1, a_2, k_1', k_2') \, ds \, du.$$

We want to shift the contours in the latter integrals, and for this we need to understand the analytic behaviour of $\mathcal{X}$. We do this in the following lemma.

First, for an L-function $L(s)$ and a positive integer a, denote by $L_a(s)$ the function formed from the same Euler product as $L(s)$ but with the primes dividing a removed. Note that for $\mathrm{Re}(s) > 0$,

$$|L_a(s)| \le \tau(a) |L(s)|. \tag{8}$$

Lemma 15. *Let $k_1, k_2 \in \mathbb{Z}$, and a_1, a_2 be positive integers such that $p | \gcd(k_1, k_2) \implies p | 2a_1a_2$. We have that*

$$\mathcal{X}_{N'}(\alpha, \beta, a_1, a_2, k_1, k_2) = \zeta_{2a_1a_2 \gcd(k_1,k_2) NN'}(\beta) \cdot L_{2a_1a_2 \gcd(k_1,k_2) NN'} \left(1+\alpha, f \otimes \chi_{(k_1,k_2)}\right) \times Z_{N'}(\alpha, \beta, a_1, a_2, k_1, k_2),$$

where $\chi_{(k_1,k_2)}$ is a primitive Dirichlet character of conductor $q | k_1^2 + k_2^2$, and $Z_{N'}$ is given by some Euler product absolutely convergent in $\mathrm{Re}(\alpha) \ge -1$ and $\mathrm{Re}(\beta) \ge 3/2 + \varepsilon$, and uniformly bounded in terms of N and N'.

Proof. From Lemma 13, we know that $G_{(\ell k_1, \ell k_2)}(n)$ is jointly multiplicative in n and ℓ, hence we can study the Euler product

$$\left(\prod_{\substack{p | N' \\ p \nmid 2a_1a_2}} \sum_{n, \ell \ge 0} \frac{\lambda_f(p^n)}{p^{\alpha n + \beta \ell}} \frac{G_{(p^\ell k_1, p^\ell k_2)}(p^{n+\nu_p(N')})}{p^{2n+2\nu_p(N')}} \right) \cdot \left(\prod_{\substack{p \nmid N' \\ p \nmid 2a_1a_2}} \sum_{n, \ell \ge 0} \frac{\lambda_f(p^n)}{p^{\alpha n + \beta \ell}} \frac{G_{(p^\ell k_1, p^\ell k_2)}(p^n)}{p^{2n}} \right).$$

For now, we focus on $p \nmid N'$. For the generic prime $p \nmid 2a_1a_2 \gcd(k_1, k_2)$, we have for $\ell = 0$ by Proposition 11 that the terms that survive are

$$1 + \frac{\lambda_f(p) \left(\frac{k_1^2 + k_2^2}{p} \right)}{p^{\alpha+1}}.$$

Since p is odd, we can also write $\left(\frac{k_1^2+k_2^2}{p} \right) = \chi_{(k_1,k_2)}(p) 1_{(k_1^2+k_2^2, p)=1}$, where $\chi_{(k_1,k_2)}(p)$ is some primitive character of conductor $q | k_1^2 + k_2^2$. For $\ell = 1$, the only terms that appear are $n = 0$ and $n = 2$, which give a contribution of

$$\frac{1}{p^\beta} + \frac{\lambda_f(p^2) \left(\frac{-1}{p} \right)}{p^{2\alpha+\beta+2}}.$$

While for $\ell = 2$, one gets

$$\frac{1}{p^{2\beta}} + \frac{\lambda_f(p^2)(p^2 - \rho(p))}{p^{2\alpha+2\beta+2}} + \frac{\lambda_f(p^3) \left(\frac{k_1^2+k_2^2}{p} \right)}{p^{3\alpha+2\beta+1}}.$$

We start detecting a pattern for even and odd ℓ's, and so we group all the terms in a way to get that the full contribution from all ℓ's and n's in the case $p \nmid 2a_1a_2 \gcd(k_1,k_2)NN'$ is

$$1+\sum_{\ell=1}^{\infty}\frac{1}{p^{\ell\beta}}+\sum_{r=1}^{\infty}\lambda_f(p^{2r})\left[\frac{\left(\frac{-1}{p}\right)}{p^{2r\alpha+(2r-1)\beta+2}}+\frac{p^2-\rho(p)}{p^{2r\alpha+2}}\sum_{\ell=2r}^{\infty}\frac{1}{p^{\ell\beta}}\right]+\left(\frac{k_1^2+k_2^2}{p}\right)\sum_{r=0}^{\infty}\frac{\lambda_f(p^{2r+1})}{p^{(2r+1)\alpha+2r\beta+1}}.$$

For $\mathrm{Re}(\alpha)\geq -1$ and $\mathrm{Re}(\beta)\geq 3/2+\varepsilon$, this is

$$\begin{aligned}&1+\sum_{\ell\geq 1}\frac{1}{p^{\beta\ell}}+\frac{\lambda_f(p)\left(\frac{k_1^2+k_2^2}{p}\right)}{p^{\alpha+1}}+O\left(\frac{1}{p^{2\,\mathrm{Re}(\alpha)+\mathrm{Re}(\beta)+2}}+\frac{1}{p^{2\,\mathrm{Re}(\alpha)+2\,\mathrm{Re}(\beta)}}\right)\\&=1+\sum_{\ell\geq 1}\frac{1}{p^{\beta\ell}}+\frac{\lambda_f(p)\left(\frac{k_1^2+k_2^2}{p}\right)}{p^{\alpha+1}}+O(p^{-(1+2\varepsilon)}).\end{aligned}$$

Now we can extract the $\zeta_{2a_1a_2\gcd(k_1,k_2)NN'}(\beta)$ and $L_{2a_1a_2\gcd(k_1,k_2)NN'}(1+\alpha, f\otimes\chi_{(k_1,k_2)})$ factors appearing in the statement of the lemma, from the first three terms of the previous line.

In the case $p|\gcd(k_1,k_2)$, we don't get a contribution since $p\nmid 2a_1a_2$ by definition of $\mathcal{X}$, and $p|\gcd(k_1,k_2)\implies p|2a_1a_2$.

In the case $p|NN'$, we get an Euler product of the form

$$\prod_{p|NN'}\left(1+O\left(p^{O(1)}\right)\right)\ll_{N,N'}1,$$

which we put in the $Z_{N'}$ factor.

□

In view of Lemma 15, we first split the sum over (k_1',k_2') into two parts $k_1'^2+k_2'^2\leq W$ and $k_1'^2+k_2'^2>W$ where $W=U^2X^{-1}Y^4$. For the former range, we shift the contour in s to $\mathrm{Re}(s)=\frac{7}{4}$. For the latter range, we shift the contour in s to $\mathrm{Re}(s)=\frac{5}{2}$. In both cases, we shift the contour in u to $\mathrm{Re}(u)=-1$.

For the range $k_1'^2+k_2'^2\leq W$, we get

$$-\frac{N'}{16\pi}\sum_{\substack{(a_1,2NN')=1\\a_1\leq Y}}\frac{\mu(a_1)}{a_1^4}\sum_{\substack{a_2|N\\(a_2,N')=1}}\frac{\mu(a_2)}{a_2^2}\sum_{\substack{(k_1',k_2')\neq(0,0)\\p|\gcd(k_1',k_2')\implies p|2a_1a_2\\k_1^2+k_2^2\leq W}}(-1)^{k_1'+k_2'}\rho_{(k_1',k_2')}(a_1^2)\rho_{(k_1',k_2')}(a_2)$$

$$\times\int_{(-1)}\int_{(7/4)}\tilde{\tilde{h}}(1-s/2,u+s)\left(\frac{a_1^2a_2N'}{\sqrt{k_1'^2+k_2'^2}}\right)^s\frac{\Gamma(s/2)}{\pi^s\cdot\Gamma(1-s/2)}\mathcal{X}_{N'}(1/2+u,s,a_1,a_2,k_1',k_2')\,ds\,du.$$

We proceed to bound this. Using (7) and (8), together with the bound $\rho(n) \leq n\tau(n)$ (Lemma 14) and the fact that $\tau(n)$ is submultiplicative, we get the bound

$$\ll_{N'} X^{1/8}U^{3/4} \sum_{\substack{(a_1,2NN')=1\\ a_1\leq Y}} \tau(a_1)^{O(1)}a_1^{3/2} \sum_{\substack{a_2|N\\ (a_2,N')=1}} \tau(a_2)^{O(1)}a_2^{3/4} \int_{(-1)}\int_{(3/2)} \frac{|\Gamma(s/2)|}{|\Gamma(1-s/2)|}$$
$$\times \sum_{\substack{(k_1',k_2')\neq(0,0)\\ p|\gcd(k_1',k_2') \Longrightarrow p|2a_1a_2\\ k_1'^2+k_2'^2\leq W}} \frac{\tau(k_1'^2+k_2'^2)^{O(1)}L(\frac{1}{2}, f\otimes\chi_{(k_1',k_2')})}{(k_1'^2+k_2'^2)^{7/8}} \frac{ds\,du}{|u|(1+|s|)^{98}(1+|u|)^{98}}.$$

We want to bound the sum inside the integral first. To do this, note that for $k_1'^2+k_2'^2 = m^2d$ where d is square-free, we have $\chi_{(k_1',k_2')}(n) = \chi_d(n)$ for $(n,m)=1$. Hence we get, by using (8) again, that

$$\sum_{\substack{(k_1',k_2')\neq(0,0)\\ p|\gcd(k_1',k_2') \Longrightarrow p|2a_1a_2\\ k_1'^2+k_2'^2\leq W}} \frac{\tau(k_1'^2+k_2'^2)^{O(1)}L(\frac{1}{2}+it, f\otimes\chi_{(k_1',k_2')})}{(k_1'^2+k_2'^2)^{7/8}}$$
$$\ll \sum_{m\leq\sqrt{W}} \frac{\tau(m)^{O(1)}}{m^{7/4}} \sum_{d\leq W/m^2} r(m^2d)\tau(m^2d)^{O(1)} \frac{|L(\frac{1}{2}+it, f\otimes\chi_d)|}{d^{7/8}}.$$

Now, using the submultiplicativity of $\tau(n)$ and the property that $r(m^2d) \leq r(m^2)r(d) \leq 4\tau(m)^2r(d)$, we obtain that the latter is bounded by

$$\ll \sum_{m\leq\sqrt{W}} \frac{\tau(m)^{O(1)}}{m^{7/4}} \sum_{d\leq W/m^2} r(d)\tau(d)^{O(1)} \frac{|L(\frac{1}{2}+it, f\otimes\chi_d)|}{d^{7/8}}.$$

We can now do a dyadic decomposition for the inner sum and then perform Cauchy-Schwarz. After this, we are left to use the work of Li [14], i.e. Lemma 7, to bound the second moment of the twisted L-function of f. Putting all of the above together, the sum over (k_1', k_2') finally gives

$$\ll_N W^{1/8}(\log X)^{O(1)}.$$

Hence, at the end, we get the bound

$$\ll_N X^{1/8}U^{3/4}W^{1/8}Y^{5/2}(\log X)^{O(1)}.$$

By a similar calculation in the range $k_1'^2+k_2'^2 > W$, we get

$$\ll_N \frac{U^{3/2}Y^4(\log X)^{O(1)}}{X^{1/4}W^{1/4}}.$$

Using the definitions of $U = \frac{X}{(\log X)^A}, Y = (\log X)^{A/10}$, and $W = U^2X^{-1}Y^4$, the off-diagonal terms give an error of size

$$\ll_N U\cdot Y^3(\log X)^{O(1)} \ll_{N,A} \frac{X}{(\log X)^{A/100}}.$$

4.2 THE $\mathcal{B}_U$ SUM

We want to estimate

$$\sum_{\substack{d\,SF\\(d,2N)=1}} r(d)\mathcal{B}_U(f\otimes\chi_{8d})\psi\left(\frac{8d}{X}\right).$$

We bound this using Cauchy-Schwarz, and then employing [25, Proposition 4.3][3], together with

$$\sum_{d\leq x} r(d)^2 \asymp x\log x,$$

to get that the sum is

$$\ll X\sqrt{\log X}(\log\log X)^3.$$

REFERENCES

1. Hung M. Bui, Alexandra Florea, Jonathan P. Keating, and Edva Roditty-Gershon, *Moments of quadratic twists of elliptic curve L-functions over function fields*, Algebra Number Theory **14** (2020), no. 7, 1853–1893. MR 4150252
2. Daniel Bump, Solomon Friedberg, and Jeffrey Hoffstein, *Nonvanishing theorems for L-functions of modular forms and their derivatives*, Invent. Math. **102** (1990), no. 3, 543–618. MR 1074487
3. John Coates, Yongxiong Li, Ye Tian, and Shuai Zhai, *Quadratic twists of elliptic curves*, Proc. Lond. Math. Soc. (3) **110** (2015), no. 2, 357–394. MR 3335282
4. I. S. Gradshteyn and I. M. Ryzhik, *Table of integrals, series, and products*, sixth ed., Academic Press, Inc., San Diego, CA, 2000, Translated from the Russian, Translation edited and with a preface by Alan Jeffrey and Daniel Zwillinger. MR 1773820
5. Benedict H. Gross and Don B. Zagier, *Heegner points and derivatives of L-series*, Invent. Math. **84** (1986), no. 2, 225–320. MR 833192
6. Tinghao Huang, *Averaging quadratically twisted l-values and their derivatives*, 2025.
7. Tinghao Huang and Zhining Wei, *First moment of derivatives of l-functions in a nonlinear family*, (2026).
8. H. Iwaniec and E. Kowalski, *Analytic number theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004. MR 2061214
9. Henryk Iwaniec, *On the order of vanishing of modular L-functions at the critical point*, Sém. Théor. Nombres Bordeaux (2) **2** (1990), no. 2, 365–376. MR 1081731
10. Yujiao Jiang, Quanli Shen, and Ziyang Tang, *The second moment of derivatives of quadratic twists of modular l-functions*, 2026.
11. W. Kohnen and D. Zagier, *Values of L-series of modular forms at the center of the critical strip*, Invent. Math. **64** (1981), no. 2, 175–198. MR 629468
12. V. A. Kolyvagin, *Finiteness of $E(\mathbb{Q})$ and Ш$(E/\mathbb{Q})$ for a subclass of Weil curves*, Izv. Akad. Nauk SSSR Ser. Mat. **52** (1988), no. 3, 522–540, 670–671. MR 954295
13. Sumit Kumar, Kummari Mallesham, Prahlad Sharma, and Saurabh K. Singh, *Moments of derivatives of modular L-functions*, Q. J. Math. **75** (2024), no. 2, 715–734. MR 4765788
14. Xiannan Li, *Moments of quadratic twists of modular L-functions*, Invent. Math. **237** (2024), no. 2, 697–733. MR 4768632
15. Ritabrata Munshi, *On mean values and non-vanishing of derivatives of L-functions in a nonlinear family*, Compos. Math. **147** (2011), no. 1, 19–34. MR 2771124
16. ———, *On quadratic families of CM elliptic curves*, Trans. Amer. Math. Soc. **363** (2011), no. 8, 4337–4358. MR 2792990
17. M. Ram Murty and V. Kumar Murty, *Mean values of derivatives of modular L-series*, Ann. of Math. (2) **133** (1991), no. 3, 447–475. MR 1109350

[3]Note that Proposition 4.3 of [25] also covers the case $f = g$ in the notation there. This is similar to the work of Li [14] where he obtained the same result but for the L-functions instead of their derivatives, see [14, Proposition 6.1].

18. Ian Petrow, *Moments of $L'(\frac{1}{2})$ in the family of quadratic twists*, Int. Math. Res. Not. IMRN (2014), no. 6, 1576–1612. MR 3180602
19. Maksym Radziwiłł and K. Soundararajan, *Moments and distribution of central L-values of quadratic twists of elliptic curves*, Invent. Math. **202** (2015), no. 3, 1029–1068. MR 3425386
20. Matthias Schütt and Tetsuji Shioda, *Mordell-Weil lattices*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics [Results in Mathematics and Related Areas. 3rd Series. A Series of Modern Surveys in Mathematics], vol. 70, Springer, Singapore, 2019. MR 3970314
21. Quanli Shen, *The first moment of quadratic twists of modular L-functions*, Acta Arith. **206** (2022), no. 4, 313–337. MR 4528422
22. K. Soundararajan and Matthew P. Young, *The second moment of quadratic twists of modular L-functions*, J. Eur. Math. Soc. (JEMS) **12** (2010), no. 5, 1097–1116. MR 2677611
23. J.-L. Waldspurger, *Sur les coefficients de Fourier des formes modulaires de poids demi-entier*, J. Math. Pures Appl. (9) **60** (1981), no. 4, 375–484. MR 646366
24. Jean-Loup Waldspurger, *Correspondances de Shimura et quaternions*, Forum Math. **3** (1991), no. 3, 219–307. MR 1103429
25. Zijie Zhou, *Moment of derivatives of quadratic twists of modular l-functions*, 2025.

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